



Fundamentals of Metal Casting

Groover M.P.: Chapters 7 and 8



The student:

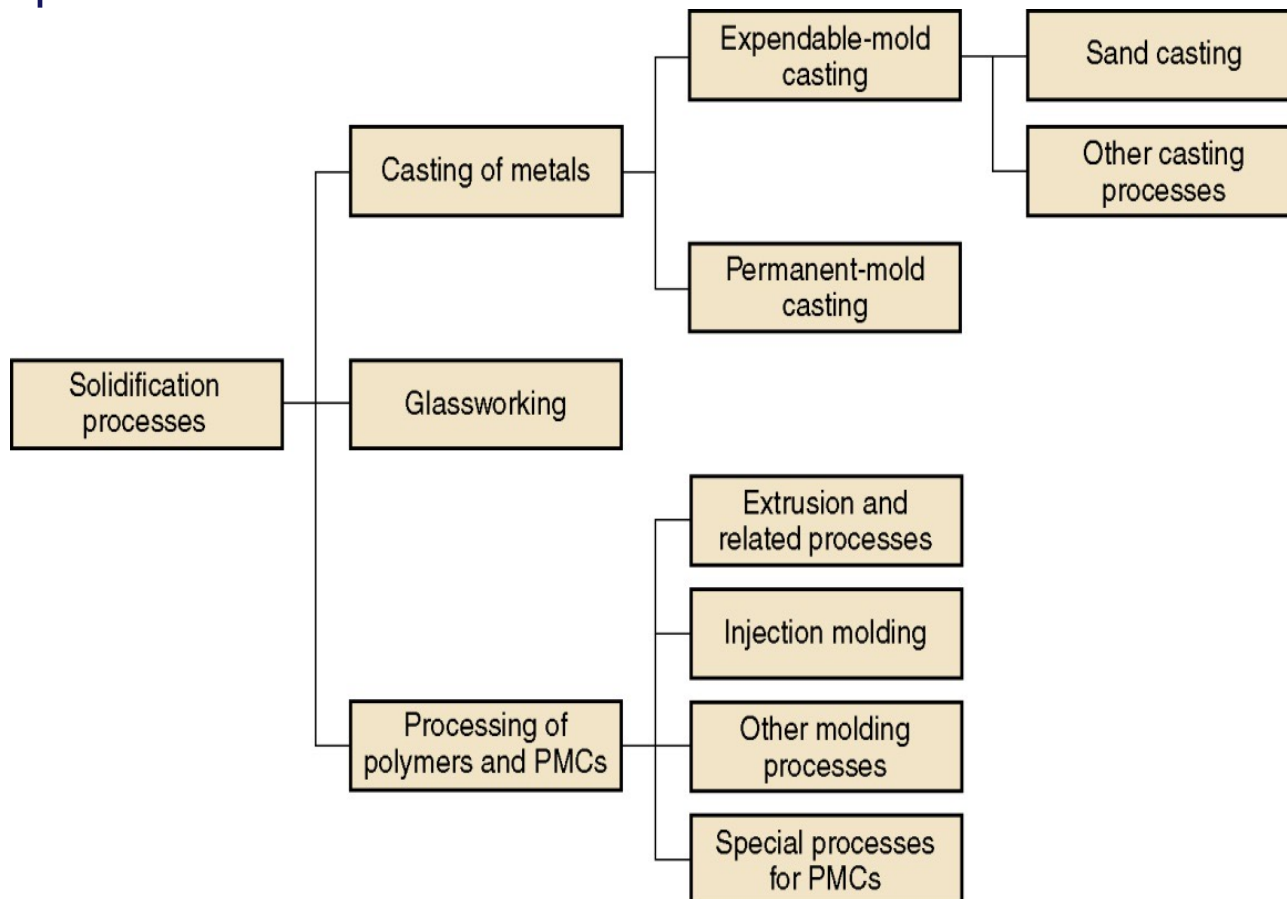
- has an overview of casting technology;
- knows about heating and pouring;
- knows about solidification and cooling;
- knows about metal casting processes.





SOLIDIFICATION PROCESSES

Starting work material is either a liquid or is in a highly plastic condition, and a part is created through solidification of the material. Solidification processes can be classified according to engineering material processed.





CASTING OF METALS

Process in which molten metal flows by gravity or other force into a mold where it solidifies in the shape of the mold cavity.

- The term *casting* also applies to the part made in the process.
- Steps in casting seem simple:
 1. Melt the metal;
 2. Pour it into a mold;
 3. Let it freeze.



CAPABILITIES OF CASTING

Advantages:

- Can create *complex part geometries*;
- Can create both *external and internal shapes*;
- Some casting processes are *net shape*, others are *near net shape*.

Disadvantages:

- Limitations on mechanical properties;
- Poor dimensional accuracy and surface finish for some processes (e.g. sand casting);
- Safety hazards to workers due to hot molten metals.



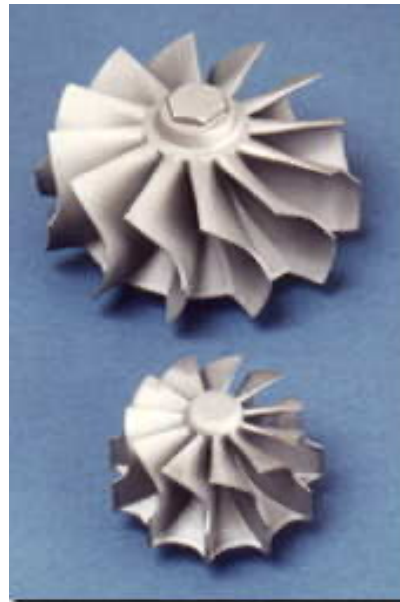
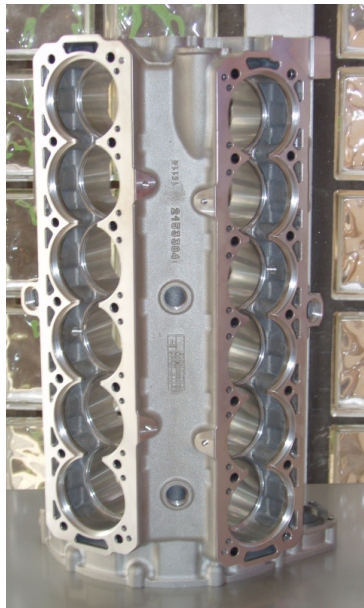
PARTS MADE BY CASTING

Big parts

- Engine blocks and heads for automotive vehicles, wood burning stoves, machine frames, railway wheels, pipes, church bells, big statues, pump housings...

Small parts

- Dental crowns, jewelry, small statues, frying pans...





OVERVIEW OF CASTING TECHNOLOGY

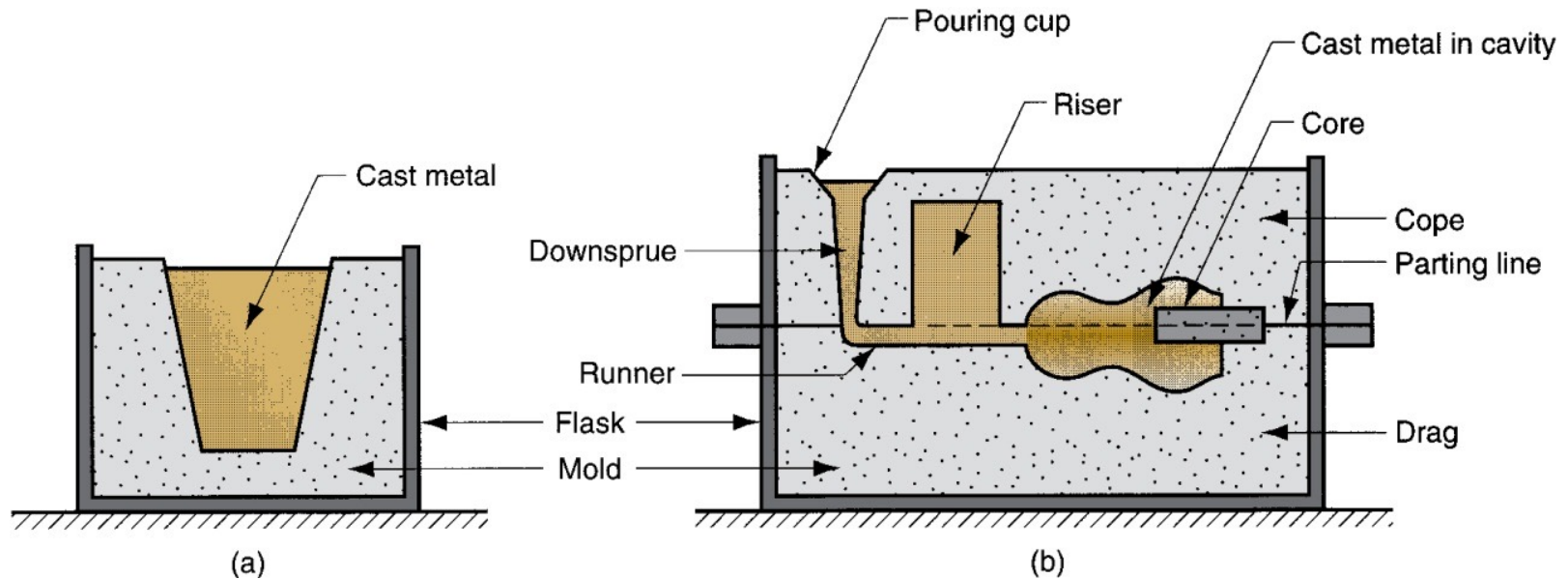
Casting is usually performed in a **foundry**, which is a factory equipped for:

- Making molds;
- Melting and handling molten metal;
- Performing the casting process;
- Cleaning the finished casting.



The mold contains cavity whose geometry determines part shape:

- Actual size and shape of cavity must be slightly enlarged to allow for shrinkage of metal during solidification and cooling;
- Molds are made of a variety of materials, including sand, plaster, ceramic, and metal.



Two forms of mold: (a) open mold and (b) closed mold for more complex mold geometry with gating system leading into the cavity.



TWO CATEGORIES OF CASTING PROCESSES

1. **Expendable mold processes** – use an expendable mold which must be destroyed to remove casting
 - Mold materials: sand, plaster, and similar materials, plus binders
2. **Permanent mold processes** – use a permanent mold which can be used to produce many castings
 - Made of metal (or, less commonly, a ceramic refractory material)

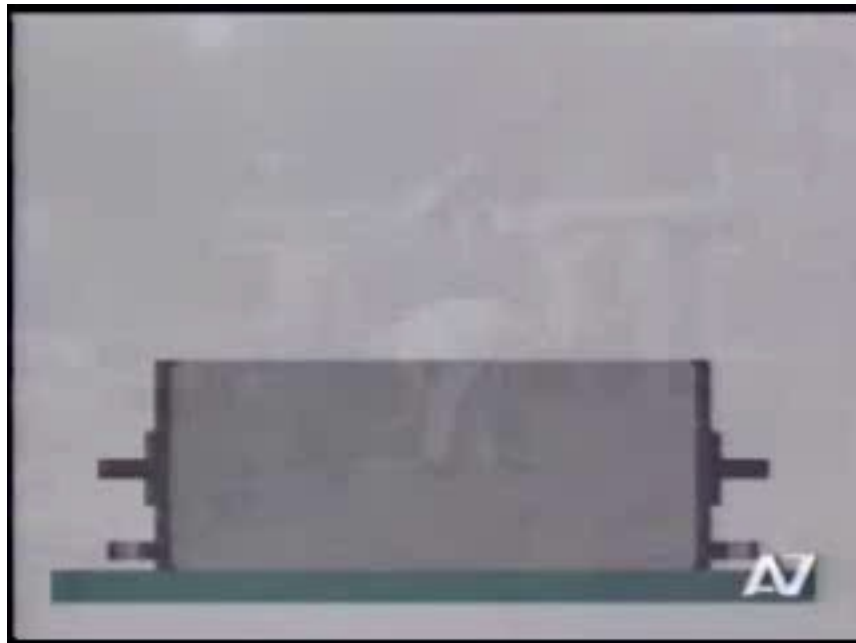
Advantages and Disadvantages:

- More intricate geometries are possible with expendable mold processes;
- Part shapes in permanent mold processes are limited by the need to open the mold;
- Permanent mold processes are more economic in high production operations.

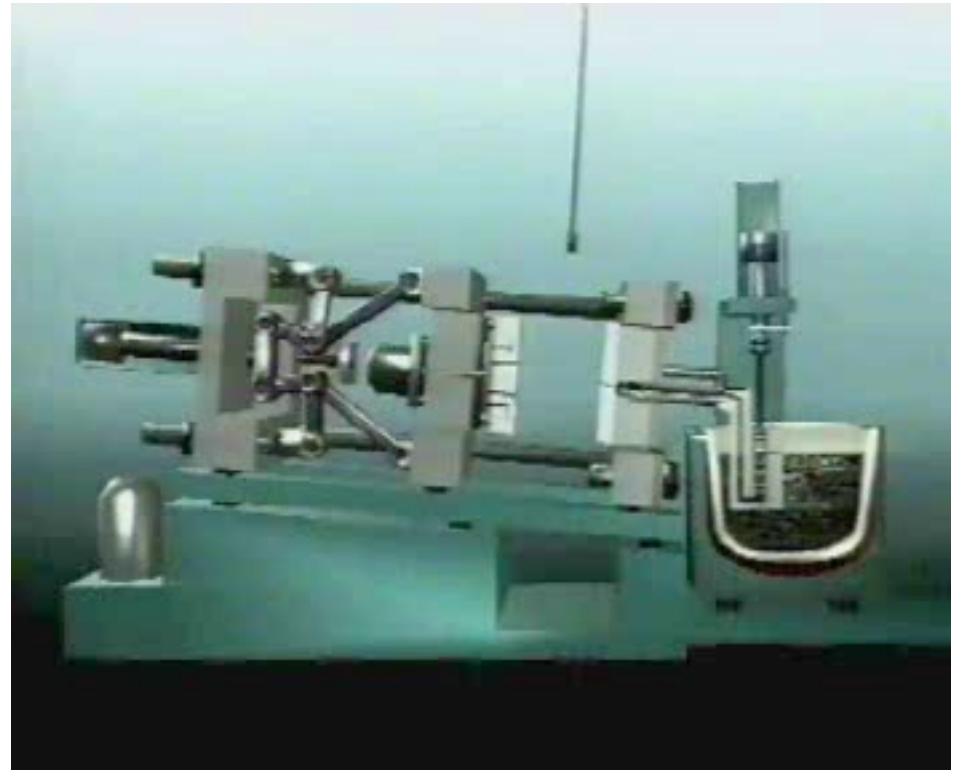


EXPENDABLE MOLD VS PERMANENT MOLD

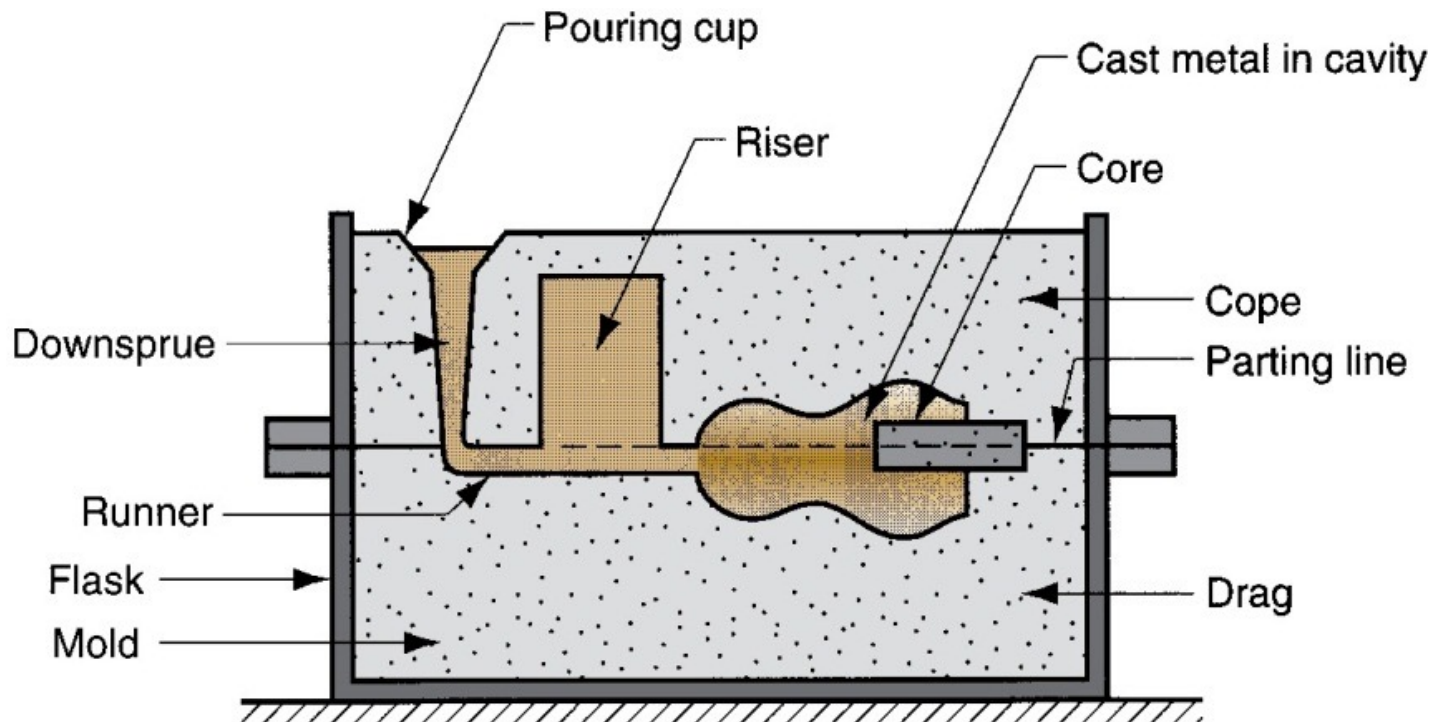
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Sand Casting



Pressure Die Casting



Mold consists of two halves:

- *Cope* = upper half;
- *Drag* = bottom half.

Mold halves are contained in a box, called a *flask*.

The two halves separate at the *parting line*.



- Mold cavity is formed by packing sand around a ***pattern***, which has the shape of the part.
- When the pattern is removed, the remaining cavity of the packed sand has desired shape of cast part.
- The pattern is usually oversized to allow for shrinkage of solidified casting during cooling to room temperature.
- Sand for the mold is moist and contains a binder to maintain its shape.
- The mold cavity provides the external surfaces of the cast part. In addition, a casting may have internal surfaces, determined by a ***core***, placed inside the mold cavity to define the interior geometry of part. In sand casting, cores are generally made of sand.



Gating System

Channels through which molten metal flows into the cavity from the outside of mold

- Consists of a *downsprue*, through which metal enters a *runner* leading to the main cavity;
- At the top of the downsprue, a *pouring cup* is often used to minimize splash and turbulence as the metal flows into the downsprue.

Riser

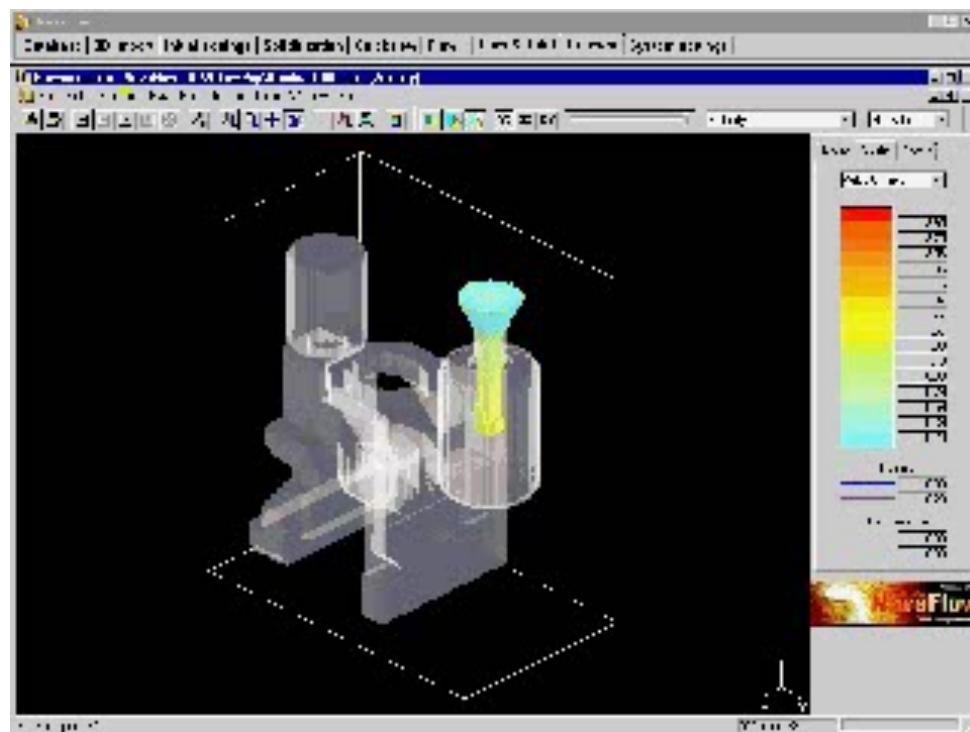
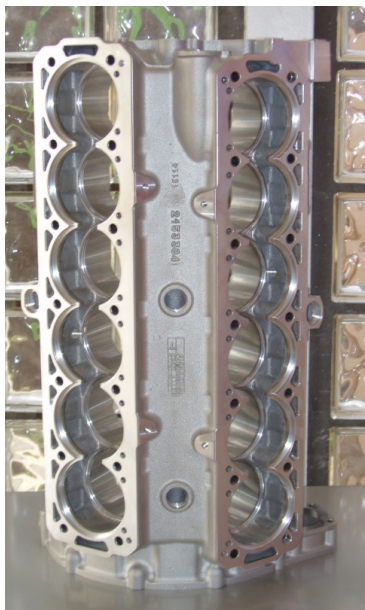
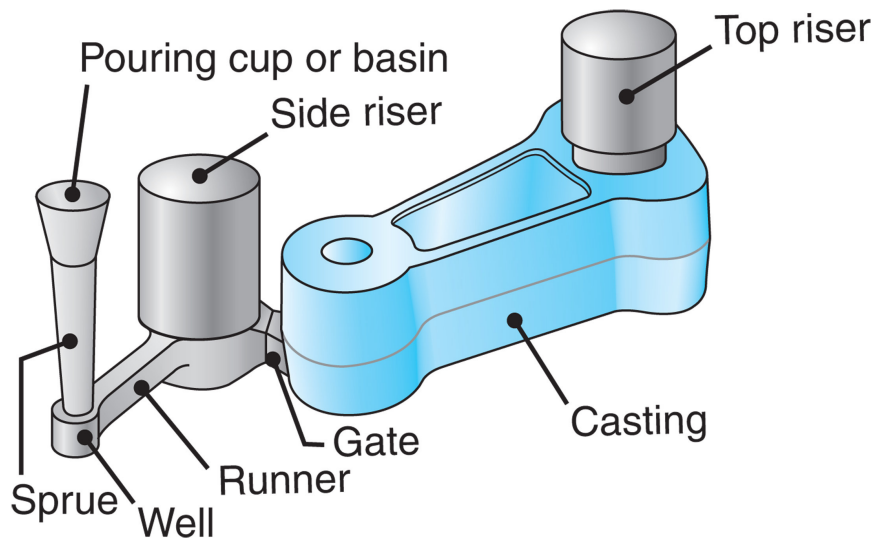
Reservoir in the mold which is a source of liquid metal to compensate for shrinkage of the part during solidification

- The riser must be designed to freeze after the main casting in order to satisfy its function.



SAND CASTING MOLD

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HEATING AND POURING THE METAL

Heating furnaces are used to heat the metal to molten temperature sufficient for casting.

The heat required is the sum of:

1. Heat to raise temperature to melting point;
2. Heat of fusion to convert from solid to liquid;
3. Heat to raise molten metal to desired temperature for pouring.

Pouring: For this step to be successful, metal must flow into all regions of the mold, especially the main cavity, before solidification begins. Factors that determine success are:

- Pouring temperature;
- Pouring rate;
- Turbulence.

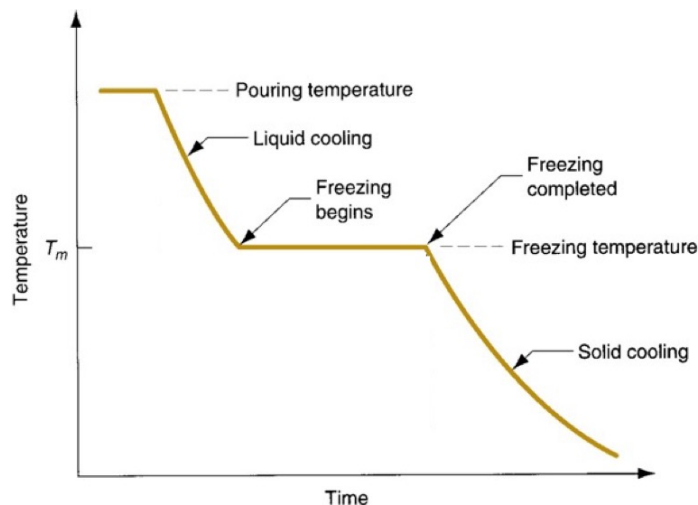


SOLIDIFICATION OF METALS

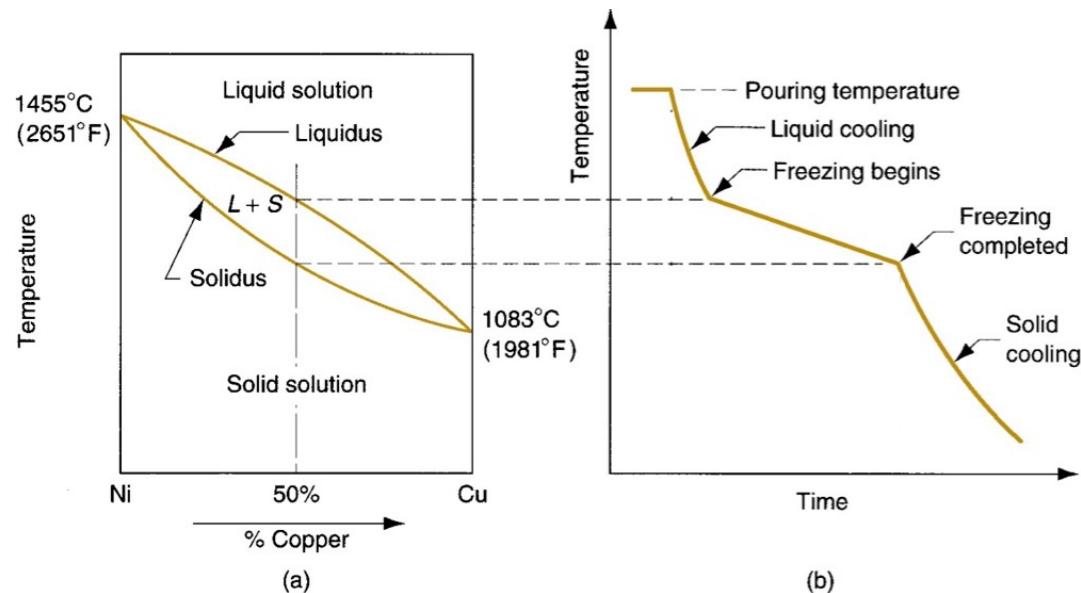
Transformation of molten metal back into solid state.

Solidification differs depending on whether the metal is (1) a pure element or (2) an alloy.

(1)



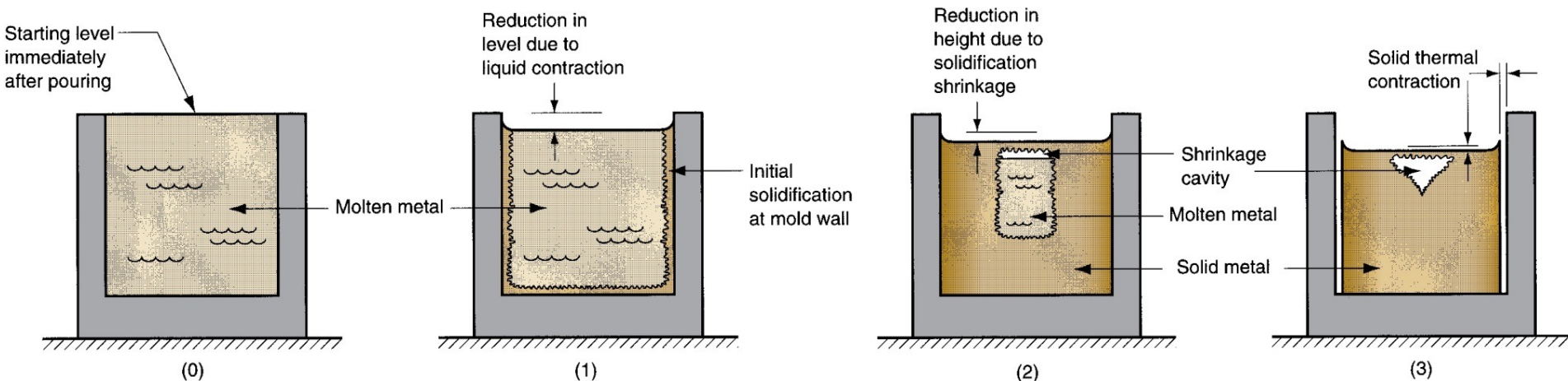
(2)





SHRINKAGE DURING SOLIDIFICATION AND COOLING

- (0) Starting level of molten metal after pouring;
- (1) Reduction in level caused by liquid contraction during cooling;
- (2) Reduction in height and formation of shrinkage cavity caused by solidification;
- (3) Further reduction in volume due to thermal contraction during cooling of solid metal.





SHRINKAGE DURING SOLIDIFICATION AND COOLING

Shrinkage occurs in nearly all metals because the solid phase has a higher density than the liquid phase.

Thus, **solidification causes a reduction in volume** per unit weight of metal, called solidification shrinkage.

Riser is the reservoir in the mold which is a source of liquid metal to **compensate for solidification shrinkage**.

Patternmakers correct for **solid thermal contraction** by making the mold cavity oversized.

Amount by which mold is made larger relative to final casting size is called ***pattern shrinkage allowance***.

Casting dimensions are expressed linearly, so allowances are applied accordingly.



Total solidification time T_{TS} is the time required for casting to solidify after pouring.

T_{TS} depends on size and shape of casting by relationship known as **Chvorinov's Rule**:

$$T_{TS} = C_m \left(\frac{V}{A} \right)^n$$

where V is the volume of the casting, A is the surface area of the casting, n is the exponent with typical value = 2, and C_m is *mold constant*.

Mold constant C_m depends on mold material, thermal properties of casting metal, and pouring temperature relative to melting point.

Value of C_m for a given casting operation can be based on experimental data from previous operations carried out using same mold material, metal, and pouring temperature, even though the shape of the part may be quite different.



WHAT CHVORINOV'S RULE TELLS US

A casting with a higher volume-to-surface area ratio cools and solidifies more slowly than one with a lower ratio:

- To feed molten metal to the main cavity, T_{TS} for riser must be greater than T_{TS} for main casting.

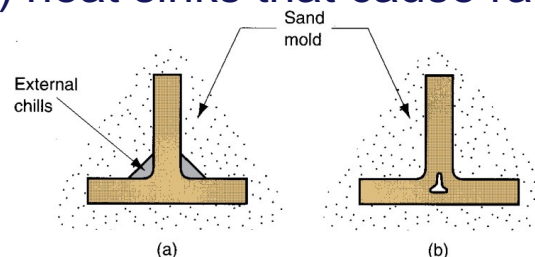
Since mold constants of riser and casting will be equal, design the riser to have a larger volume-to-area ratio so that the main casting solidifies first: This mitigates the effects of shrinkage.

To minimize effects of shrinkage, it is desirable for regions of the casting most distant from the liquid metal supply to freeze first and for solidification to progress from these regions toward the riser(s):

- Thus, molten metal is continually available from risers to prevent shrinkage voids;
- The term ***directional solidification*** describes this aspect of freezing and methods by which it is controlled.

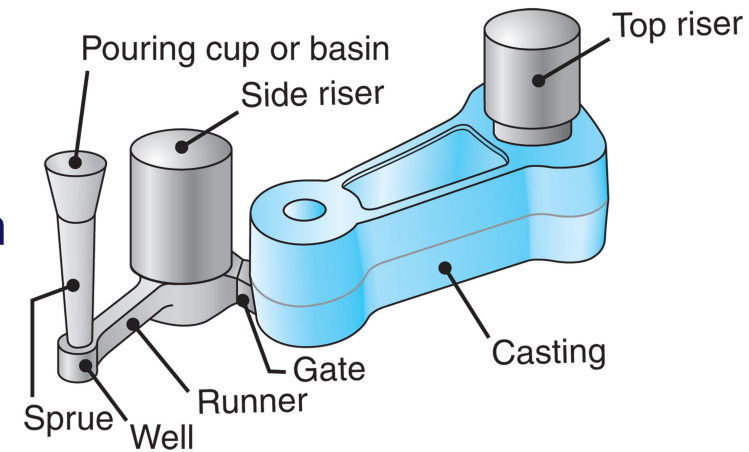
Directional solidification is achieved using Chvorinov's Rule to design the casting, its orientation in the mold, and the riser system that feeds it:

- Locate sections of the casting with lower V/A ratios away from riser, so freezing occurs first in these regions, and the liquid metal supply for the rest of the casting remains open.
- **Chills** - internal (a) or external (b) heat sinks that cause rapid freezing in certain regions of the casting.





The molten metal is poured through the gating system (consisting of a pouring basin or cup, a sprue, runners, and gates) into the mold cavity. The sprue is a tapered vertical channel through which the molten metal flows downward in the mold. Runners are the channels that carry the molten metal from the sprue to the gates. The gates are the channels that carry the molten metal from the runners into the mold cavity.

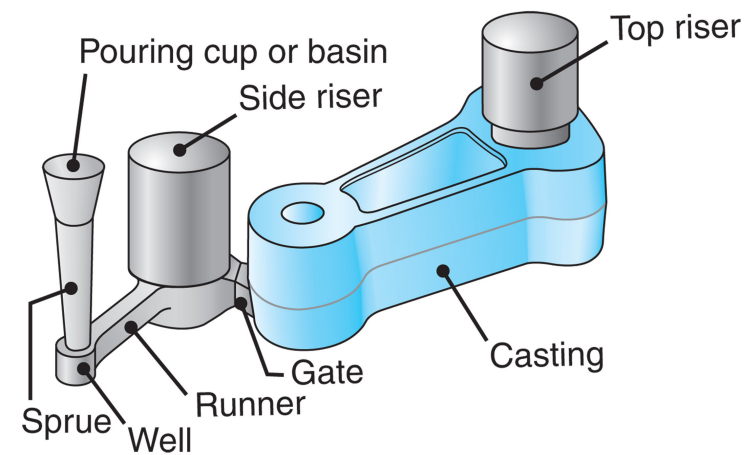
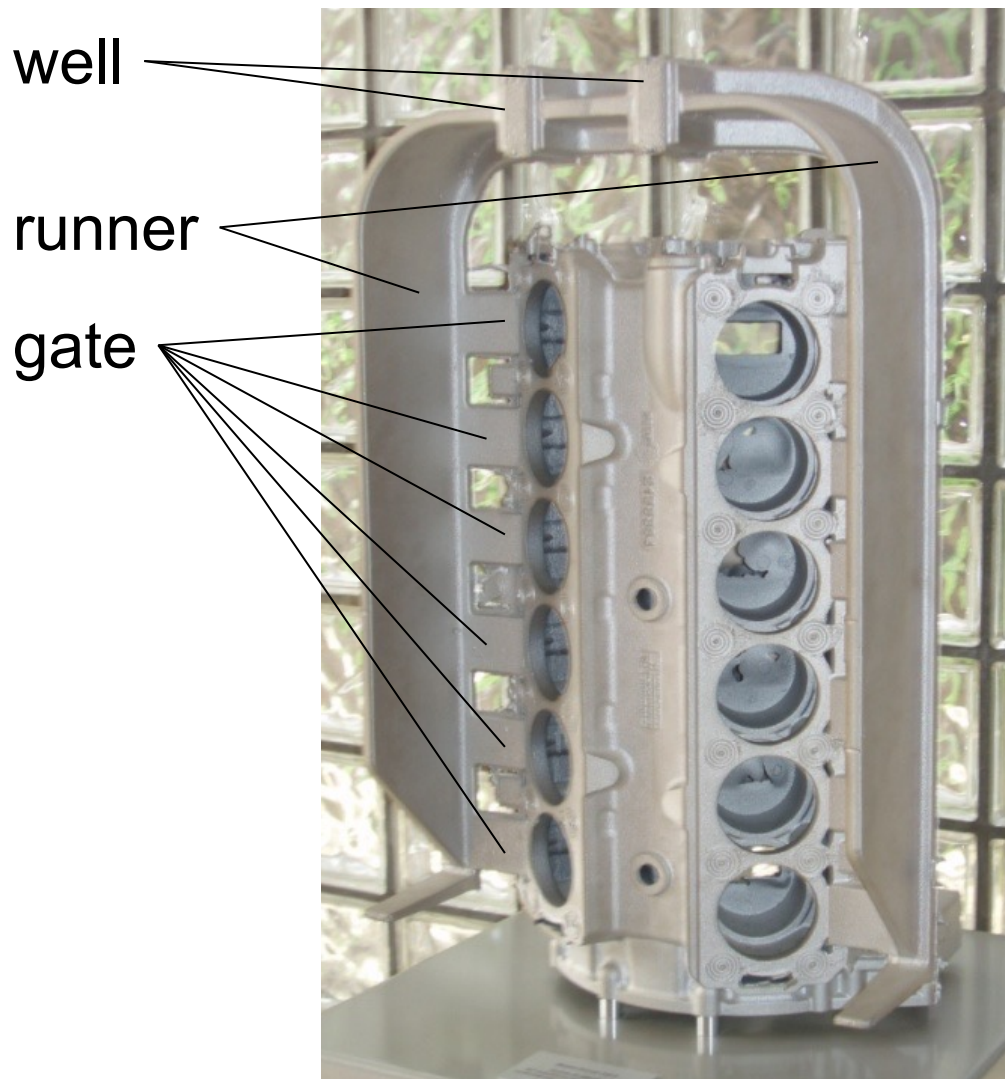


A properly designed gating system helps avoid or minimize problems such as **premature cooling**, **turbulence**, and **gas entrapment**.

The molten metal must be handled carefully to avoid the formation of oxides on molten-metal surfaces from exposure to the environment or the introduction of impurities into the molten metal.

Two basic principles of fluid flow are relevant to gating design:

1. Bernoulli's theorem, and
2. the law of mass continuity.





GATING SYSTEM – BERNOULLI'S THEOREM

This theorem is based on the principle of the **conservation of energy** and relates pressure, velocity, the elevation of the fluid at any location in the system, and the frictional losses in a system that is full of liquid. The Bernoulli's equation is:

$$h + v^2/(2g) + p/(\rho g) = \text{constant}$$

where h is the elevation above a certain reference level, p is the pressure at that elevation, v is the velocity of the liquid at that elevation, ρ is the density of the fluid (assuming that it is incompressible), and g is the gravitational constant.

Conservation of energy requires that the following relationship be satisfied:

$$h_1 + v_1^2/(2g) + p_1/(\rho g) = h_2 + v_2^2/(2g) + p_2/(\rho g) + f$$

where the subscripts 1 and 2 represent two different locations in the system and f represents the frictional loss in the liquid as it travels through the system.



Pressure can be considered constant: $p_1 = p_2$

If the pouring basin has a much larger cross sectional area than the sprue bottom, then the velocity of the molten metal at the top of the pouring basin is very low and can be taken to be zero.

If frictional losses are due to a viscous dissipation of energy, then f can be taken to be a function of the vertical distance and is often approximated as a linear function.

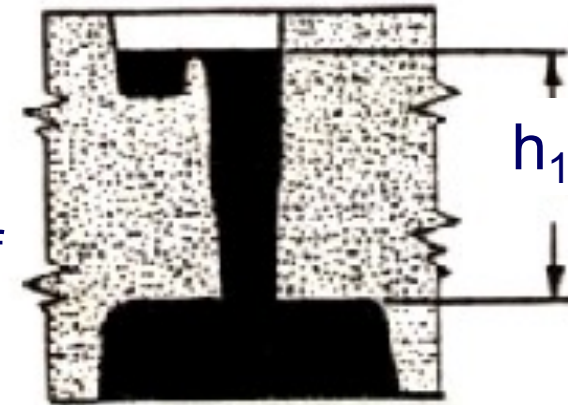
The equation becomes:

$$h_1 + p_1/(\rho g) = v_2^2/(2g) + p_1/(\rho g) + kh_1$$

$$v_2^2 = 2g(1 - k)h_1$$

The velocity v of the molten metal is: $v = c\sqrt{2gh_1}$
where c is a friction factor, always between 0 and 1.

Top gating system





GATING SYSTEM – MASS CONTINUITY LAW

The **law of mass continuity** states that, for incompressible liquids and in a system with impermeable walls, the rate of flow is constant. Thus,

$$Q = A_1 v_1 = A_2 v_2$$

where Q is the volumetric flow rate (m^3/s); v is liquid velocity; A is cross-sectional area of the liquid stream; and the subscripts refer to any two different location in the system.

Thus, an increase in area results in a decrease in velocity, and vice versa.

As the metal accelerates during its descent into the sprue opening, the cross-sectional area of the channel must be reduced; otherwise, as the velocity of the flowing metal increases toward the base of the sprue, air can be aspirated into the liquid and conducted into the mold cavity.

To prevent this condition, the sprue is designed with a taper, so that the volume flow rate is the same at the top and bottom of the sprue.



GATING SYSTEM – MASS CONTINUITY LAW

We can estimate the time required to fill a mold cavity of volume V as

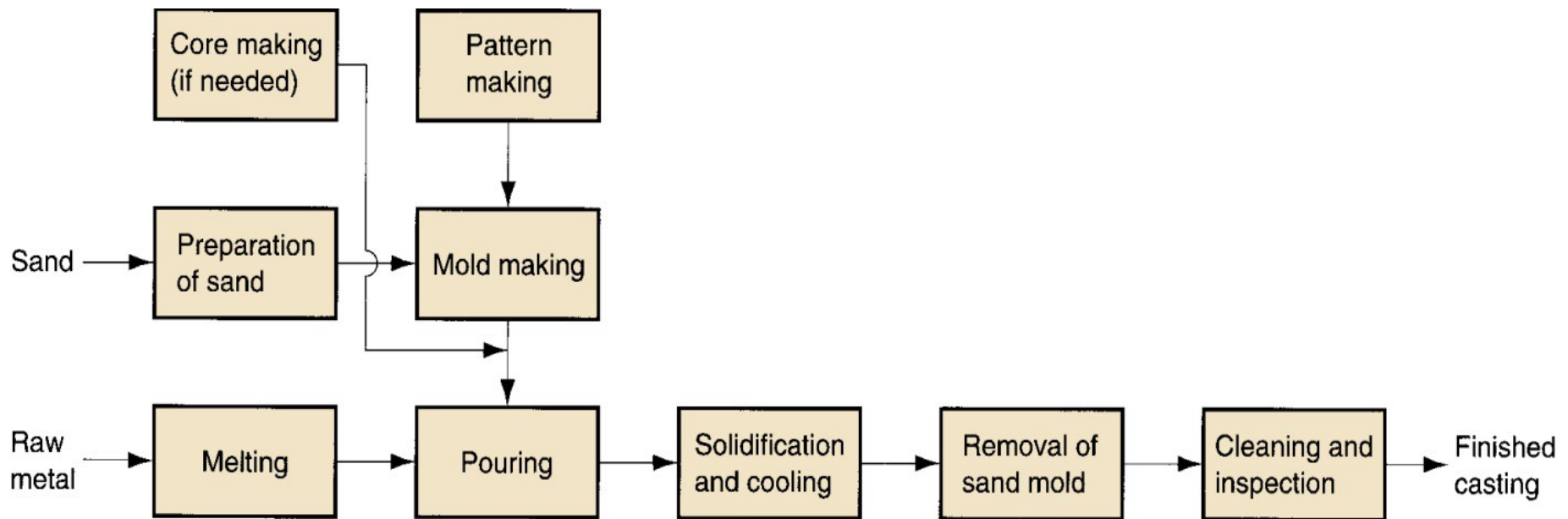
$$T_{MF} = V / Q$$

where T_{MF} is the *mold filling time* (s); V is the volume of mold cavity (m^3); and Q is the volume flow rate (m^3/s).

The mold filling time must be less than the time needed to start the solidification after the pouring, but the velocity should not be so high to erode the mold material or to generate turbulence and associated air entrainment.



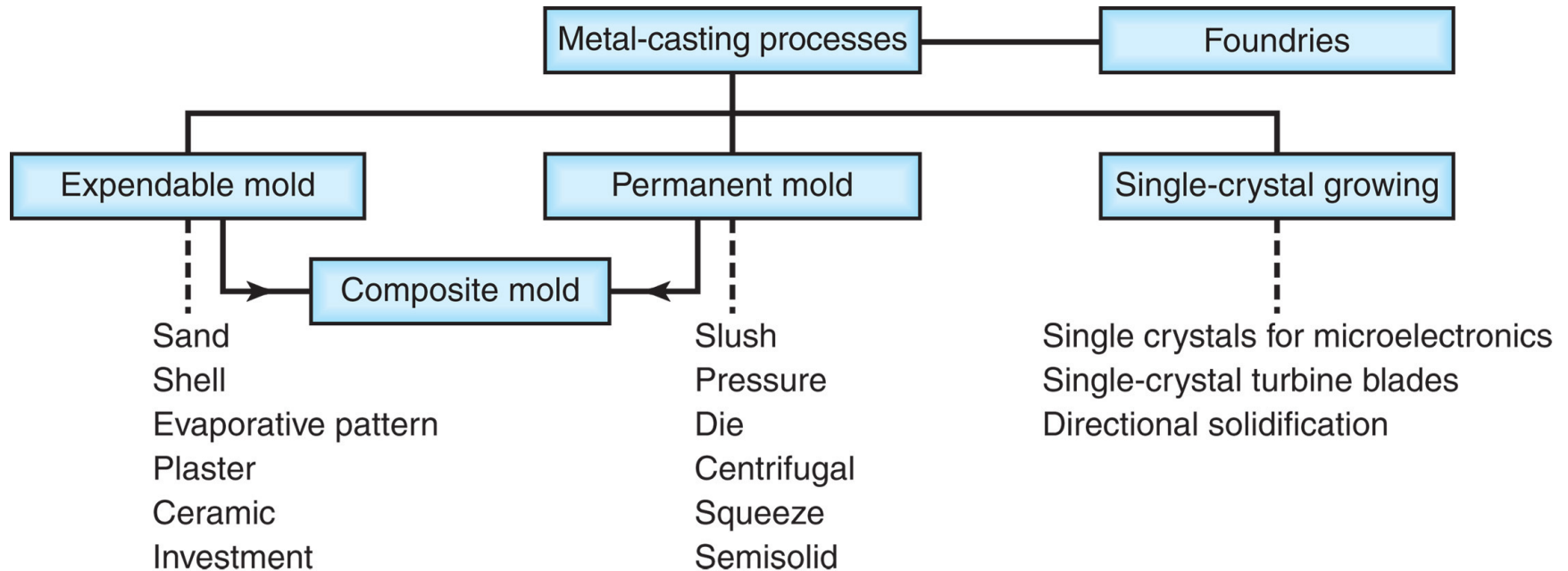
Production sequence in sand casting, including pattern-making and mold-making.





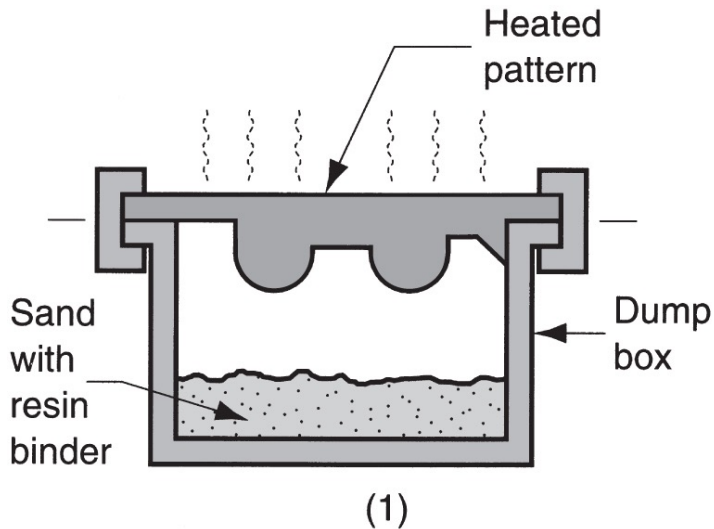


METAL CASTING PROCESSES

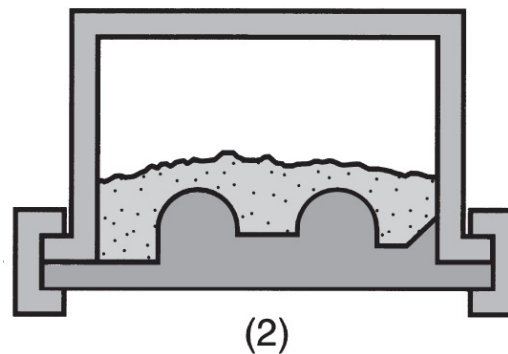




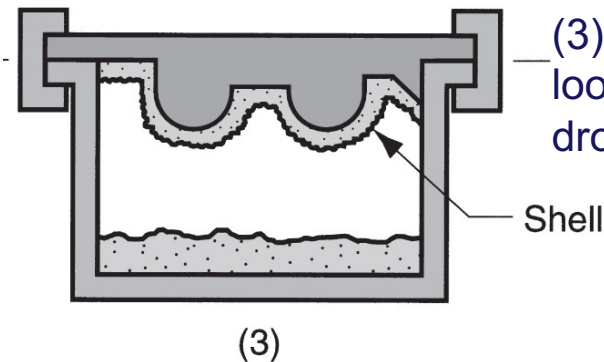
Casting process in which the mold is a thin shell of sand held together by thermosetting resin



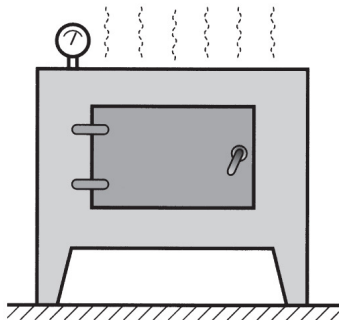
(1) A metal pattern is heated and placed over a box containing sand mixed with TS resin



(2) Box is inverted so that sand and resin fall onto the hot pattern, causing a layer of the mixture to partially cure on the surface to form a hard shell

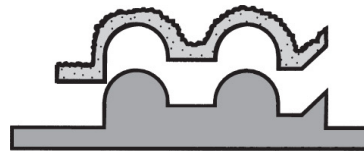


(3) Box is repositioned so loose uncured particles drop away



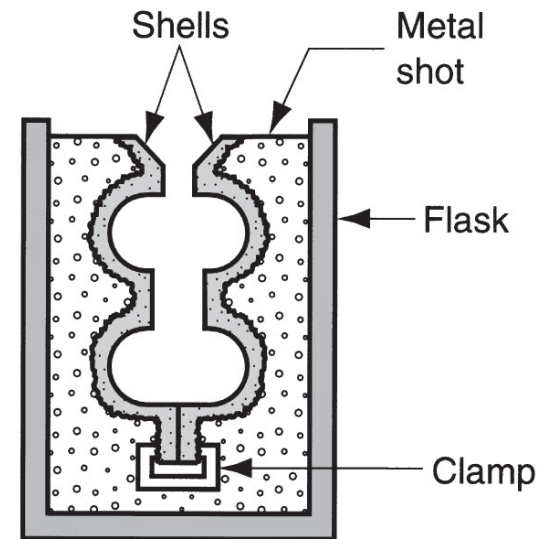
(4)

(4) Sand shell is heated in oven for several minutes to complete curing



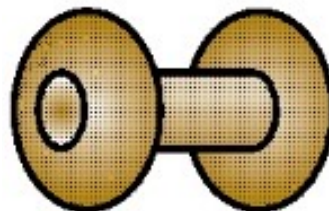
(5)

(5) shell mold is stripped from pattern



(6)

(6) Two halves of the shell mold are assembled, supported by sand or metal shot in a box, and pouring is accomplished



(7) Finished casting with sprue removed



Advantages:

- Smoother cavity surface permits easier flow of molten metal and better surface finish
- Good dimensional accuracy
- Mold collapsibility minimizes cracks in casting
- Can be automatized for mass production

Disadvantages:

- More expensive metal pattern
- Difficult to justify for small quantities



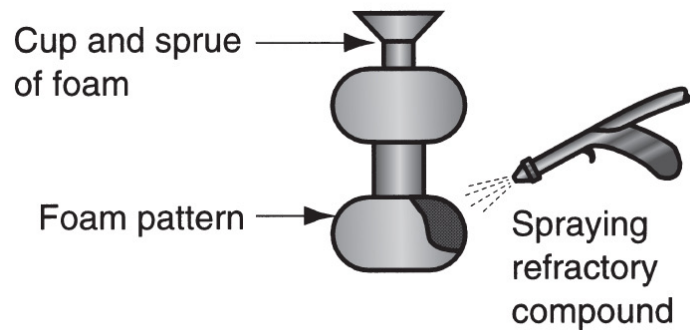
EXPANDED POLYSTYRENE PROCESS

Uses a mold of sand packed around a polystyrene foam pattern which vaporizes when molten metal is poured into mold.

- Other names: lost-foam process, lost pattern process, evaporative-foam process, and full-mold process
- Polystyrene foam pattern includes sprue, risers, gating system, and internal cores (if needed)
- Mold does not have to be opened into cope and drag sections

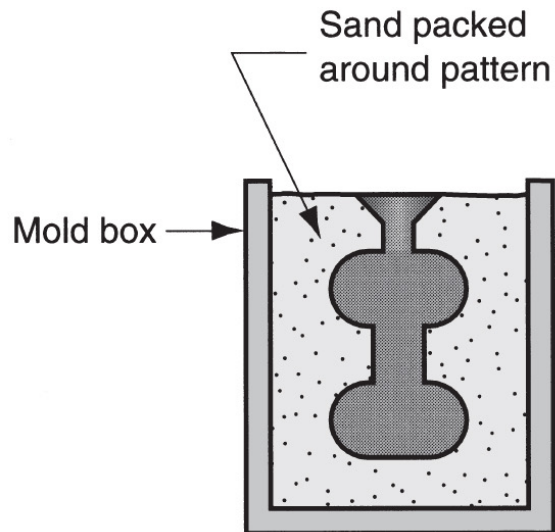


STEPS IN EXPANDED POLYSTYRENE PROCESS



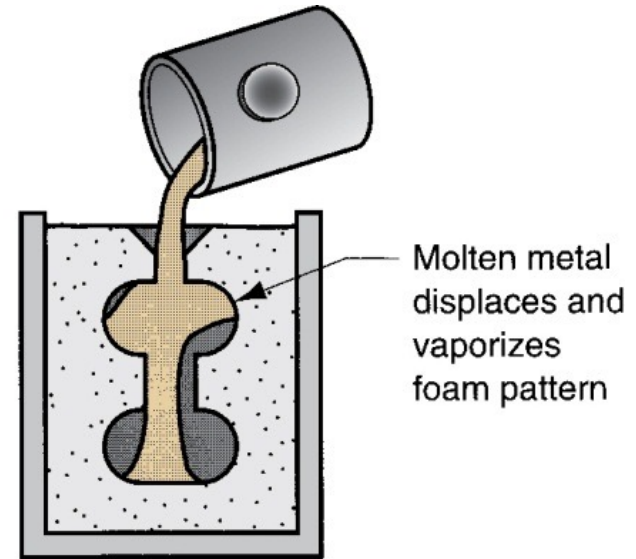
(1) Polystyrene foam pattern is coated with refractory compound

(1)



(2) Foam pattern is placed in mold box, and sand is compacted around the pattern

(2)



(3) Molten metal is poured into the portion of the pattern that forms the pouring cup and sprue

As the metal enters the mold, the polystyrene foam is vaporized ahead of the advancing liquid, thus filling the mold cavity



EXPANDED POLYSTYRENE PROCESS

Advantages of expanded polystyrene process:

- Pattern need not be removed from the mold
- Simplifies and speeds mold-making, because two mold halves are not required as in a conventional green-sand mold

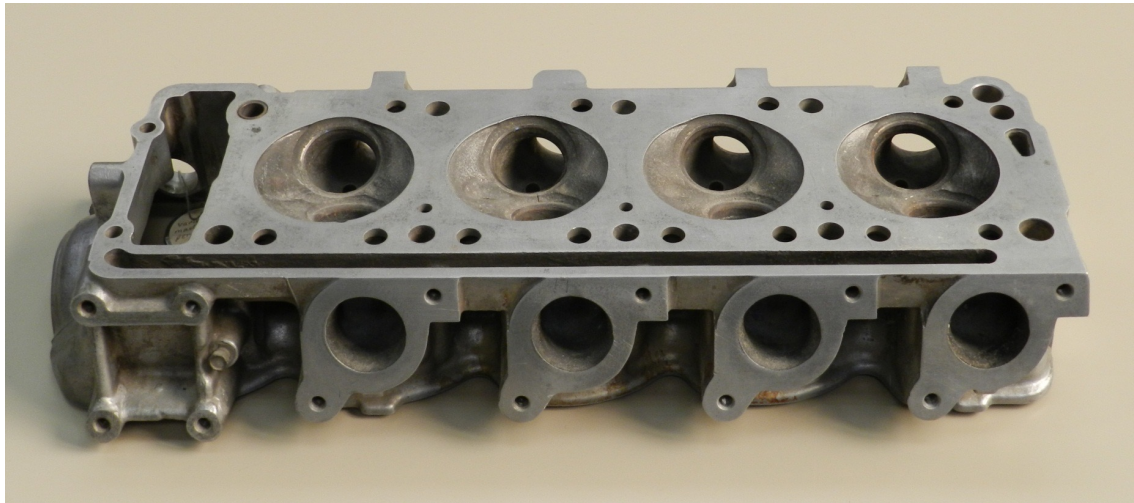
Disadvantages:

- A new pattern is needed for every casting
- Economic justification of the process is highly dependent on cost of producing patterns

EXPANDED POLYSTYRENE PROCESS

Applications:

- Mass production of castings for automobile engines
- Automated and integrated manufacturing systems are used to
 1. Mold the polystyrene foam patterns and then
 2. Feed them to the downstream casting operation



Aluminum engine head



INVESTMENT CASTING (A.K.A. LOST WAX PROCESS)

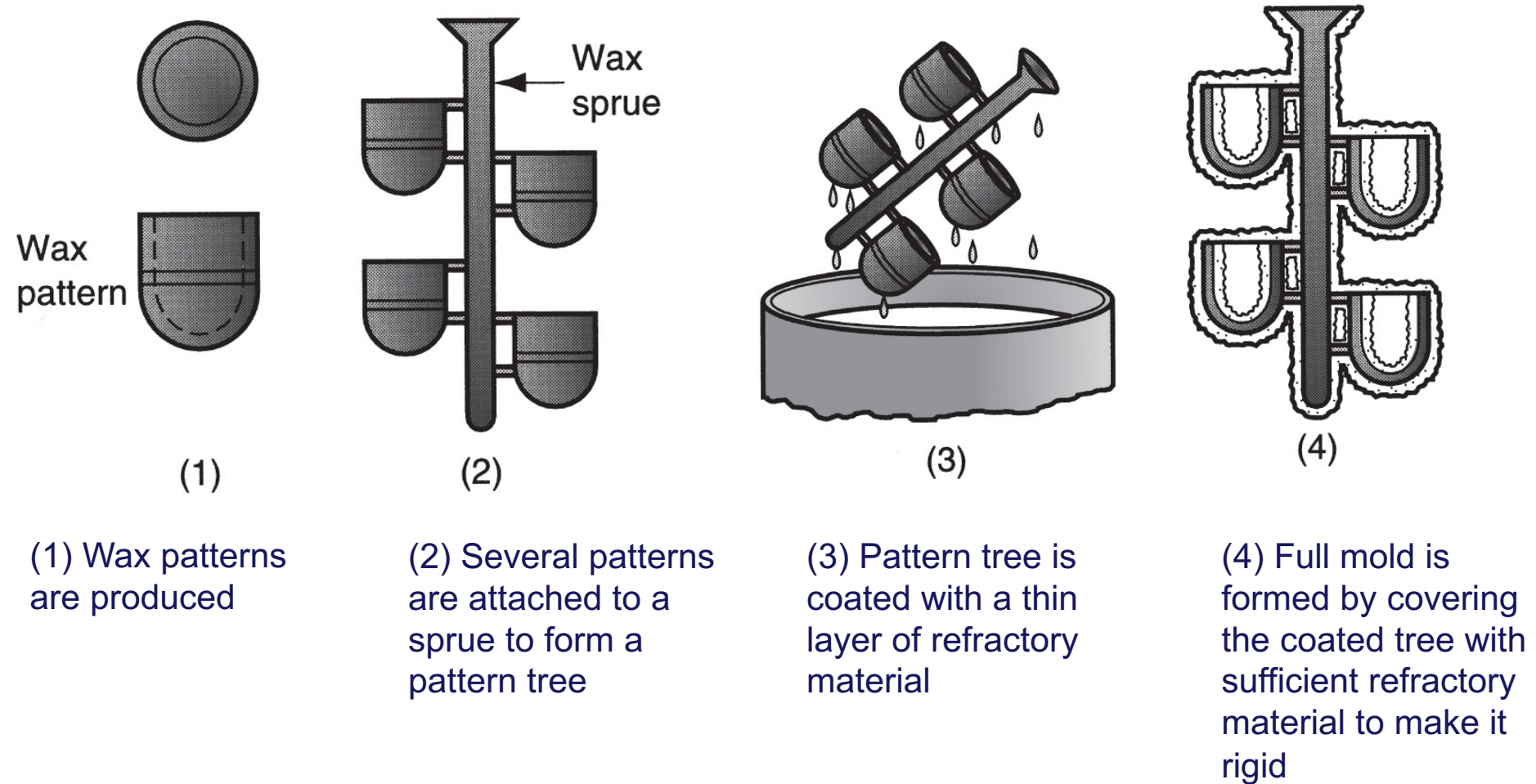
A pattern made of wax is coated with a refractory material to make the mold, after which wax is melted away prior to pouring molten metal

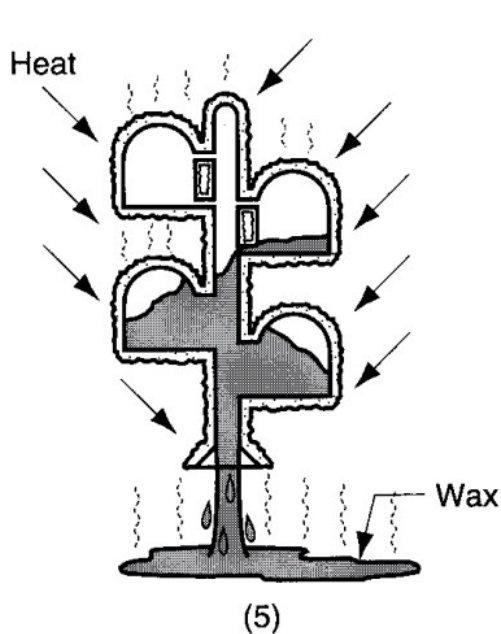
- "Investment" comes from a less familiar definition of "invest" - "to cover completely," which refers to coating of refractory material around wax pattern
- It is a precision casting process: capable of producing castings of high accuracy and intricate detail



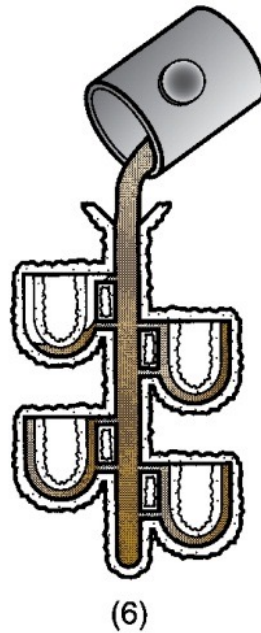
STEPS IN INVESTMENT CASTING

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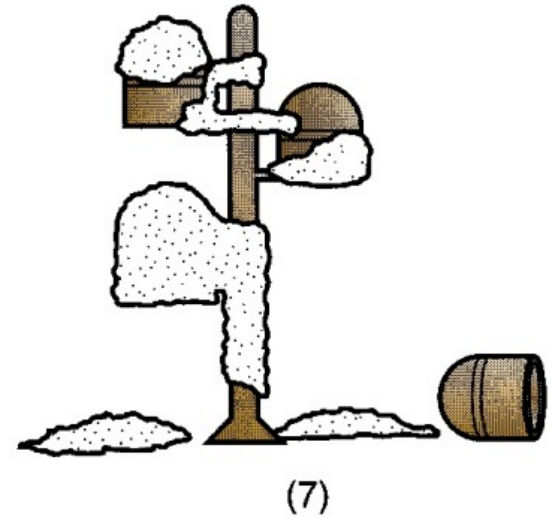




(5) Mold is held in an inverted position and heated to melt the wax and permit it to drip out of the cavity



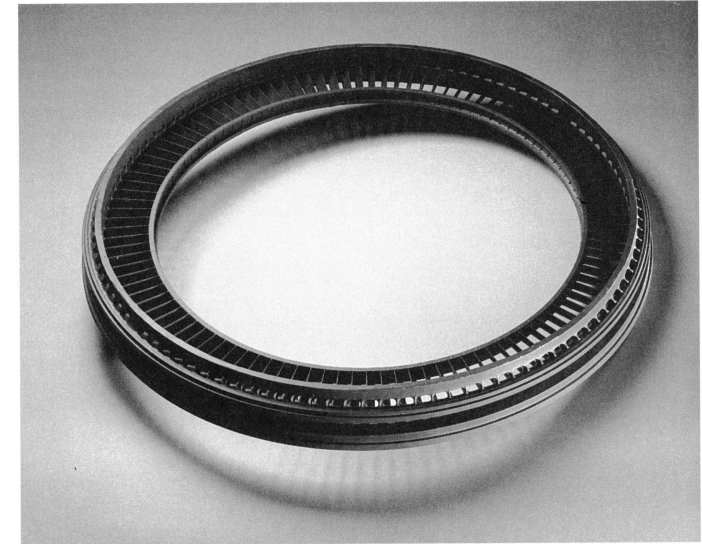
(6) Mold is preheated to a high temperature, the molten metal is poured, and it solidifies



(7) Mold is broken away from the finished casting and the parts are separated from the sprue

Advantages:

- Parts of great complexity and intricacy can be cast
- Close dimensional control and good surface finish
- Wax can usually be recovered for reuse
- This is a net shape process - additional machining is not normally required



Disadvantages:

- Many processing steps are required
- Relatively expensive process



INVESTMENT CASTING





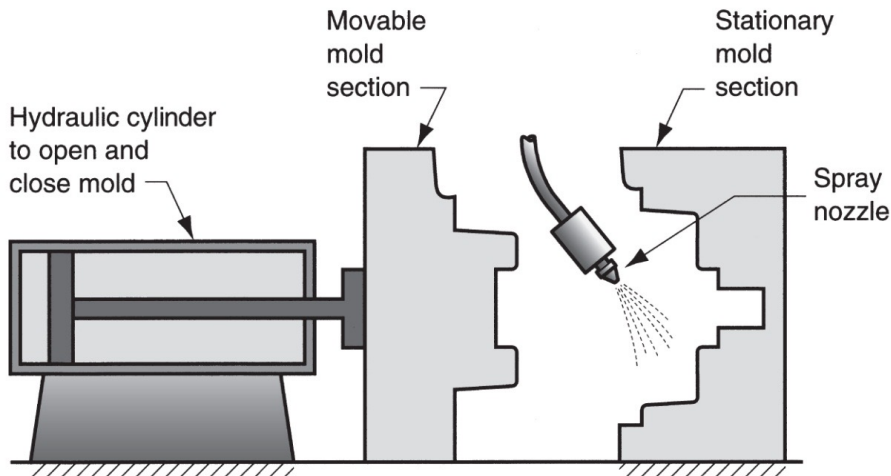
PERMANENT MOLD PROCESS

Uses a metal mold constructed of two sections designed for easy, precise opening and closing

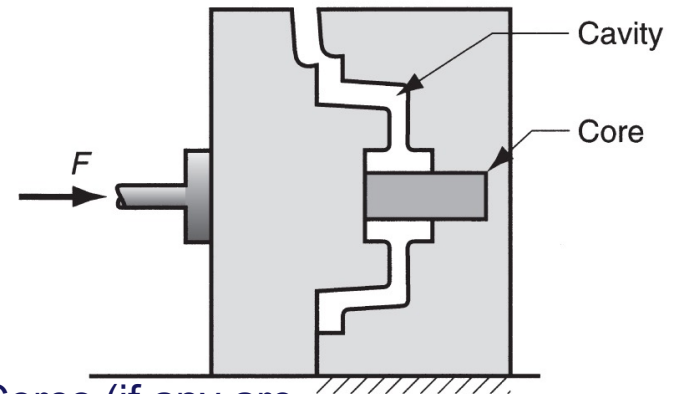
- Molds used for casting lower melting point alloys are commonly made of steel or cast iron
- Molds used for casting steel must be made of refractory material, due to the very high pouring temperatures



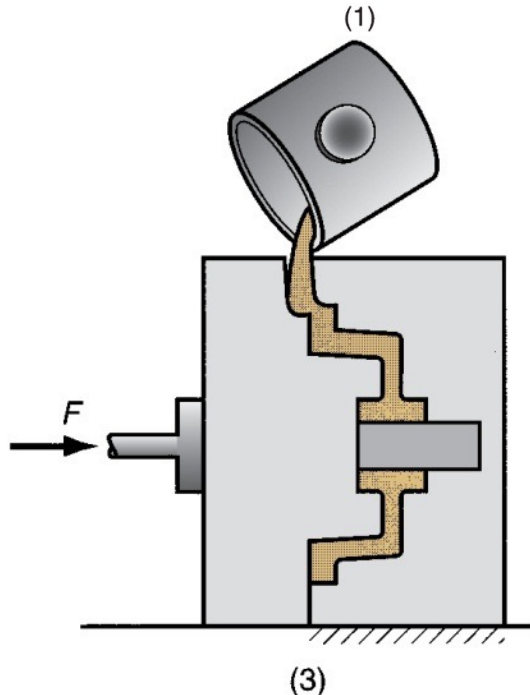
STEPS IN PERMANENT MOLD CASTING



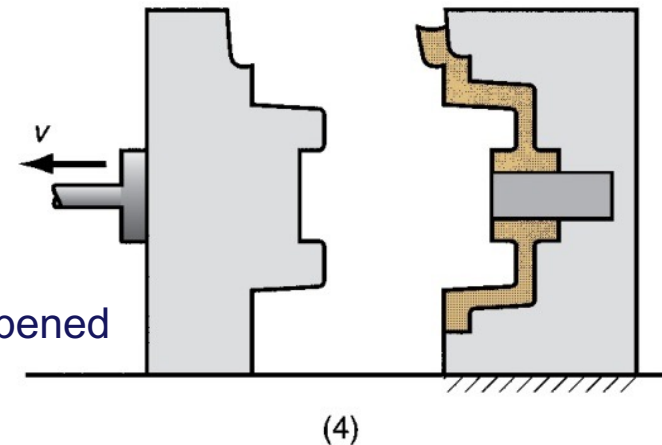
(1) Mold is preheated and coated for lubrication and heat dissipation



(2) Cores (if any are used) are inserted and mold is closed



(3) Molten metal is poured into the mold, where it solidifies



(4) Mold is opened



PERMANENT MOLD CASTING

Advantages of permanent mold casting:

- Good dimensional control and surface finish
- Rapid solidification caused by metal mold results in a finer grain structure, so castings are stronger

Limitations:

- Generally limited to metals of lower melting point
- Simpler part geometries compared to sand casting because of need to open the mold
- High cost of mold

Due to high mold cost, process is best suited to high volume production and can be automated accordingly. Typical parts: automotive pistons, pump bodies, and certain castings for aircraft and missiles

Metals commonly cast: aluminum, magnesium, copper-base alloys, and cast iron. Unsuitable to steels because of very high pouring temperatures.



A permanent mold casting process in which molten metal is injected into mold cavity under high pressure. Pressure is maintained during solidification, then mold is opened, and part is removed. Molds in this casting operation are called dies; hence the name die casting.

Use of high pressure to force metal into die cavity is what distinguishes this from other permanent mold processes.

Die casting machines are designed to hold and accurately close two mold halves and keep them closed while liquid metal is forced into cavity.

Two main types:

1. Hot-chamber machine
2. Cold-chamber machine



HOT-CHAMBER DIE CASTING

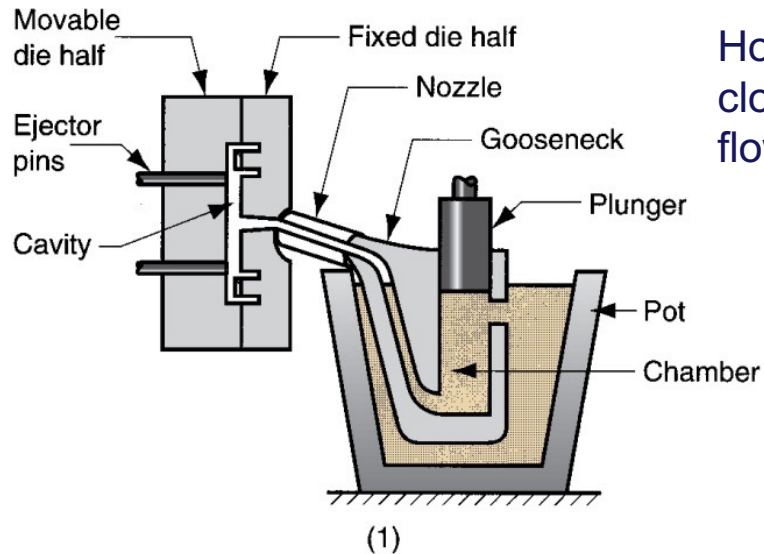
Metal is melted in a container, and a piston injects liquid metal under high pressure into the die.

- High production rates
 - 500 parts per hour not uncommon
- Applications limited to low melting-point metals that do not chemically attack plunger and other mechanical components
- Casting metals: zinc, tin, lead, and magnesium



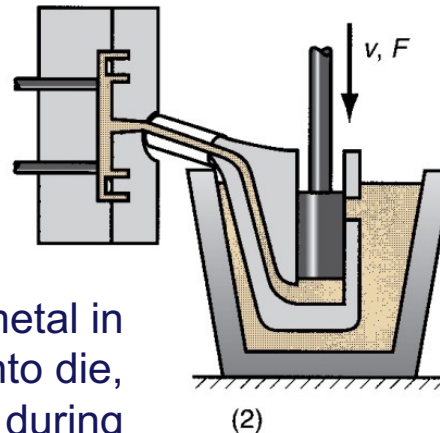
HOT-CHAMBER DIE CASTING

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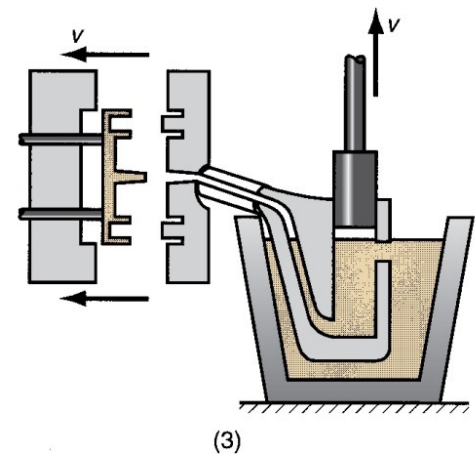


Hot-chamber die casting cycle: (1) with die closed and plunger withdrawn, molten metal flows into the chamber

(2) plunger forces metal in chamber to flow into die, maintaining pressure during cooling and solidification.



(3) Plunger is withdrawn, die is opened, and casting is ejected





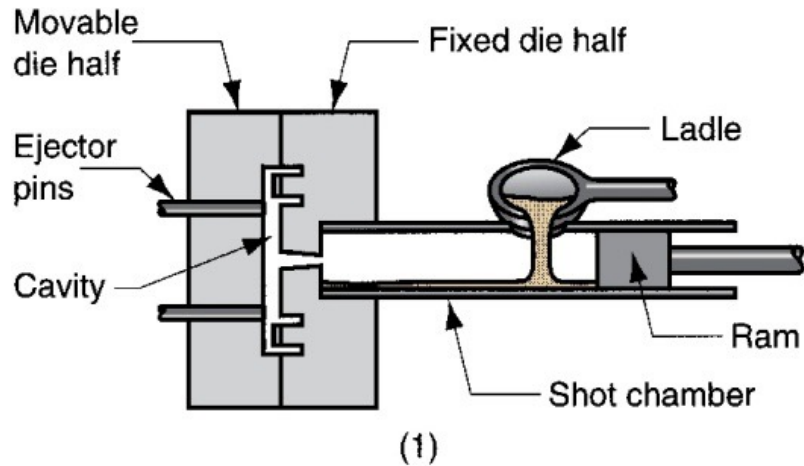
COLD-CHAMBER DIE CASTING MACHINE

Molten metal is poured into unheated chamber from external melting container, and a piston injects metal under high pressure into die cavity.

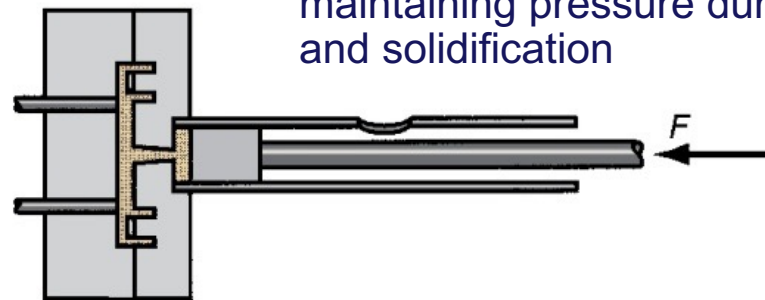
- High production but not usually as fast as hot-chamber machines because of pouring step
- Casting metals: aluminum, brass, and magnesium alloys
- Advantages of hot-chamber process favor its use on low melting-point alloys (zinc, tin, lead)



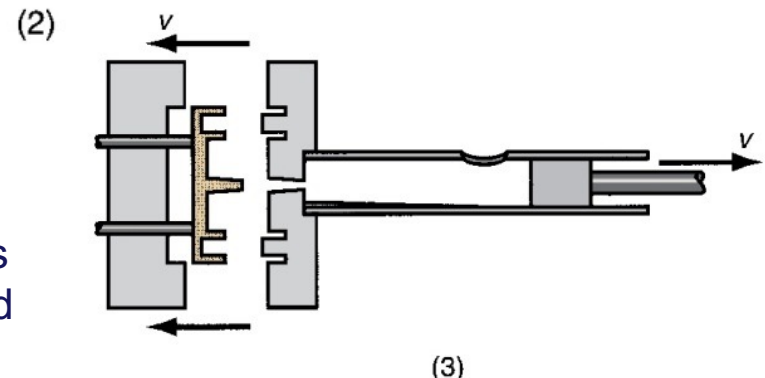
COLD-CHAMBER DIE CASTING CYCLE



(1) With die closed and ram withdrawn, molten metal is poured into the chamber



(2) Ram forces metal to flow into die, maintaining pressure during cooling and solidification



3) Ram is withdrawn, die is opened, and part is ejected



DIE CASTING: ADVANTAGES AND LIMITATIONS

Advantages:

- Economical for large production quantities
- Good accuracy and surface finish
- Thin sections possible
- Rapid cooling means small grain size and good strength in casting

Disadvantages:

- Generally limited to metals with low melting points
- Part geometry must allow removal from die



CENTRIFUGAL CASTING

A family of casting processes in which the mold is rotated at high speed so centrifugal force distributes molten metal to outer regions of die cavity. The group includes:

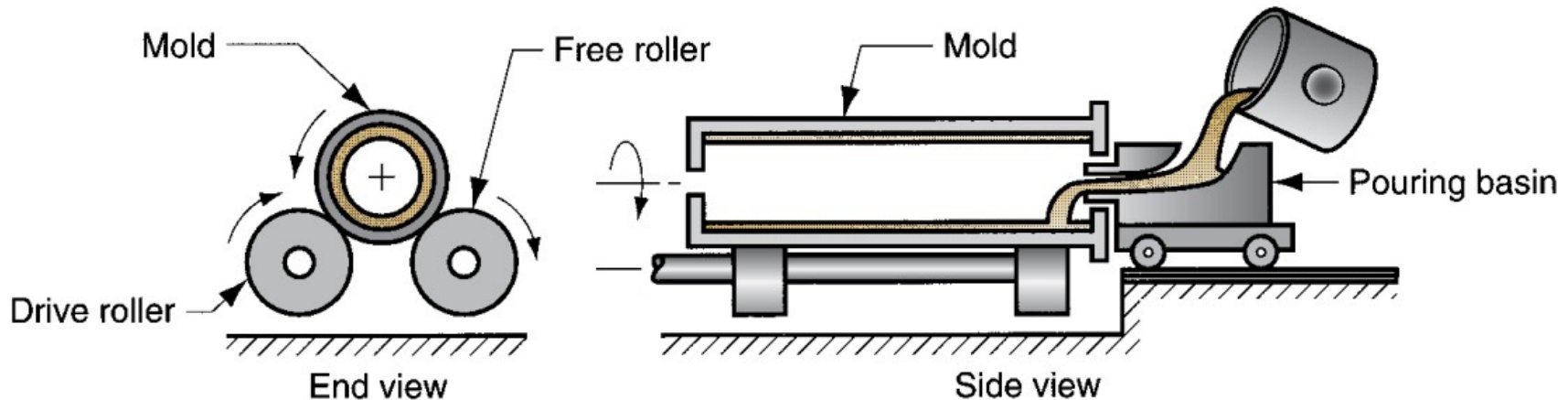
- True centrifugal casting
- Semicentrifugal casting
- Centrifuge casting



TRUE CENTRIFUGAL CASTING

Molten metal is poured into rotating mold to produce a tubular part

- In some operations, mold rotation commences after pouring rather than before
- Parts: pipes, tubes, bushings, and rings
- Outside shape of casting can be round, octagonal, hexagonal, etc., but inside shape is (theoretically) perfectly round, due to radially symmetric forces

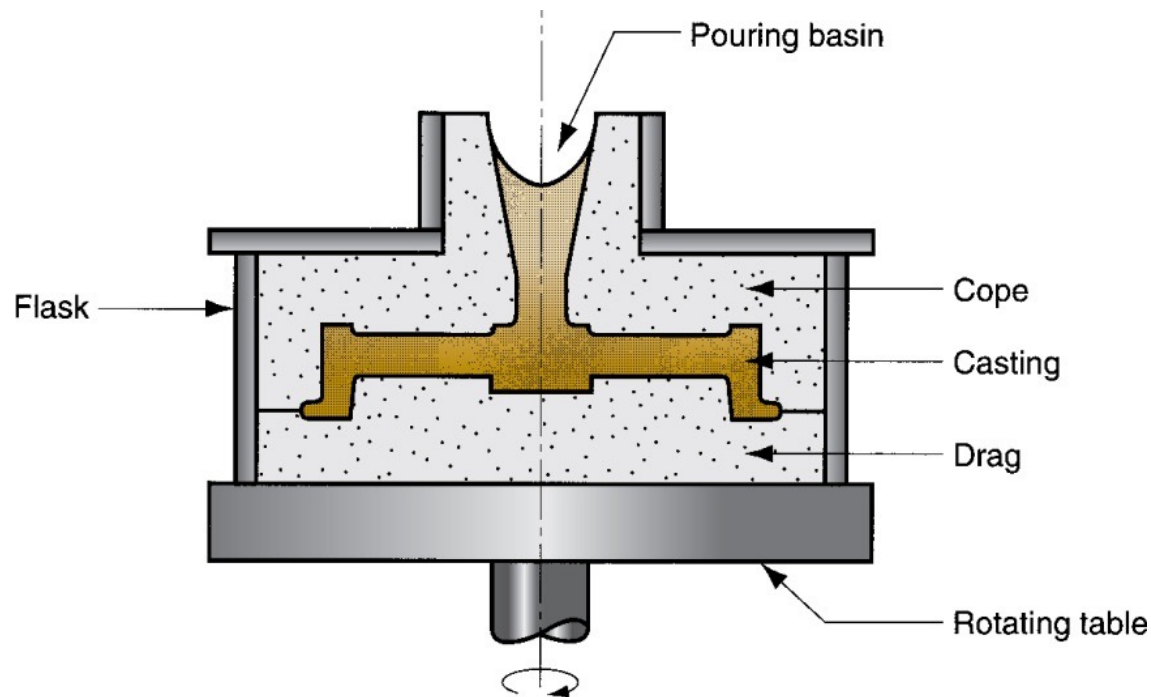




SEMICENTRIFUGAL CASTING

Centrifugal force is used to produce solid castings rather than tubular parts

- Molds use risers at center to supply feed metal
- Density of metal in final casting is greater in outer sections than at center of rotation
- Often used on parts in which center of casting is machined away, thus eliminating the portion where quality is lowest

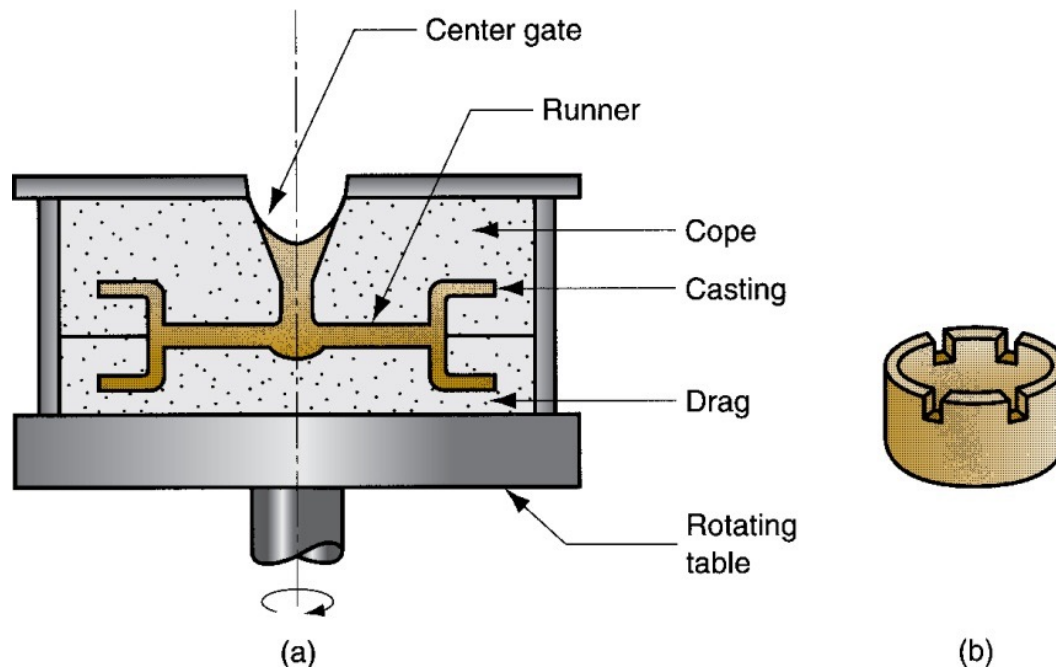




CENTRIFUGE CASTING

Mold is designed with part cavities located away from axis of rotation, so molten metal poured into mold is distributed to these cavities by centrifugal force

- Used for smaller parts
- Radial symmetry of part is not required as in other centrifugal casting methods





- Casting can create complex part geometries and (near) net shape parts, from big to small one.
- The mold contains cavity whose geometry determines part shape: it can be expendable or permanent mold.
- In sand casting pattern and cores are used to form an expendable mold. Pattern is oversized to allow for shrinkage of solid metal during cooling.
- The riser system is a source of liquid metal to compensate for shrinkage of the metal during solidification.
- Total solidification time can be estimated by the Chvorinov's Rule. The casting, its orientation in the mold, and the riser system that feeds it must be designed to achieve directional solidification.
- Gating system is constituted by channels through which molten metal flows into the cavity from the outside of mold.
- Two basic principles of fluid flow are relevant to gating design: Bernoulli's theorem, and the law of mass continuity.